Ganesh Arivoli

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Summary

Graduate student in Mechanical Engineering focused on developing GPU-accelerated physics engines for robotics and autonomous systems. Experienced in building high-fidelity simulation frameworks for accurate and fast simulation of complex systems. Skilled in C++, CUDA, and Python (incl. Warp, Taichi) with foundations in multi-body dynamics, numerical methods, and high-performance computing. Active contributor to Project Chrono, working on a scalable simulation engine for flexible multibody systems.

Education

University of Wisconsin-Madison — Aug 2024 – Dec 2026
Master of Science in Mechanical Engineering, Advisor: Dan Negrut — *Current GPA: 4.0/4.0*

National Institute of Technology, Tiruchirappalli — Jul 2017 – May 2021
Bachelor of Technology, Mechanical Engineering

Research Experience

Graduate Research Assistant | *Simulation-Based Engineering Lab, UW–Madison* — Aug 2024 – Present

- Contributed to the development of a GPU-based physics simulation engine for deformable multibody systems with support for hyperelastic materials using Total-Lagrangian FEA framework (Submitted to ***IMSD 2026***, Abstract: here).
- Developed an explicit solver and implemented optimizations to second-order solver achieving **10X speedup** over CPU baseline.
- Developed a CNN-based reduced-order model for autonomous terrain leveling with optimal control, with pipelines for training and deployment on HPC clusters, resulting in **48%** performance improvement (Published in ***Advanced Robotics Research 2026***, here).
- Evaluated GPU-based linear solvers for multibody dynamics, benchmarking against CPU-based solvers to quantify throughput and accuracy, observing a **3.6X speedup** for large FEA systems (Presented at ***ASME IDETC-CIE 2025***, Abstract: here).

Vehicle Dynamics Engineer (R&D) | *Bajaj Auto, India* — Sep 2021 – Dec 2023

- Led the design and development of suspension and steering systems for Bajaj Auto-Triumph motorcycles integrating physics-based simulation models into product development cycle, contributing to **100,000+** units sold across **58+** countries (Product site: here).
- Improved **ride and handling performance by 15%** through multibody dynamics simulations, optimizing suspension parameters.

Projects

Physics Engine for Rigid/Flexible Multibody Systems | *UW–Madison* — Aug 2025 – Nov 2025

- Developed a high-fidelity multibody dynamics simulator (rigid and flexible) from scratch using a generalized r-p formulation with constraints and BDF time-stepping for numerical integration with a custom Newton-Raphson solver.
- Extended the engine to support multi-DOF serial robotic manipulators using minimal coordinates formulation with joint-space constraints, enabling prototyping of control and trajectory planning algorithms with real-time visualization.

GPU Accelerated Path Planning for Autonomous Robots | *UW–Madison* — Jan 2025 – Apr 2025

- Implemented parallelized RRT algorithm using CUDA and OpenMP targeting computationally intensive nearest-neighbor searches and collision detection operations for planning in obstacle-dense environments.
- Prototyped kernels using PyCUDA for rapid iteration, profiled and optimized GPU bottlenecks with NVIDIA Nsight Systems, and achieved **5X speedup** over the CPU baseline (GitHub repository: here).

DEM Simulation for Komatsu Loader | *Komatsu Mining Corp, Longview, TX* — Jan 2025 – June 2025

- Developed a multibody dynamics model for a Komatsu loader and integrated it with GPU-accelerated DEM engine, enabling high-fidelity simulations of loader-granular material interaction to optimize the bucket geometry and motion (Animation: here).

SAE BAJA Racing Team | *Collegiate Team, NIT Trichy* — May 2018 – May 2021

- Led the powertrain and data acquisition teams for an All-Terrain Vehicle, earning top-5 finishes in BAJA SAE India 2020 and 2021.
- Developed and validated MATLAB-based CVT model for tuning shift logic, improving efficiency by 10% and halved tuning time.

Skills

Programming: C/C++, CUDA, Python (incl. PyCUDA, Warp, PyTorch), MATLAB, Shell (Bash, Zsh)
Tools & Platforms: Git/GitHub, Linux, NVIDIA Nsight, Docker, CMake, Bazel, ROS
Simulation Tools: Project Chrono, Mujoco, NVIDIA Isaac Sim, Simulink, MSC Adams